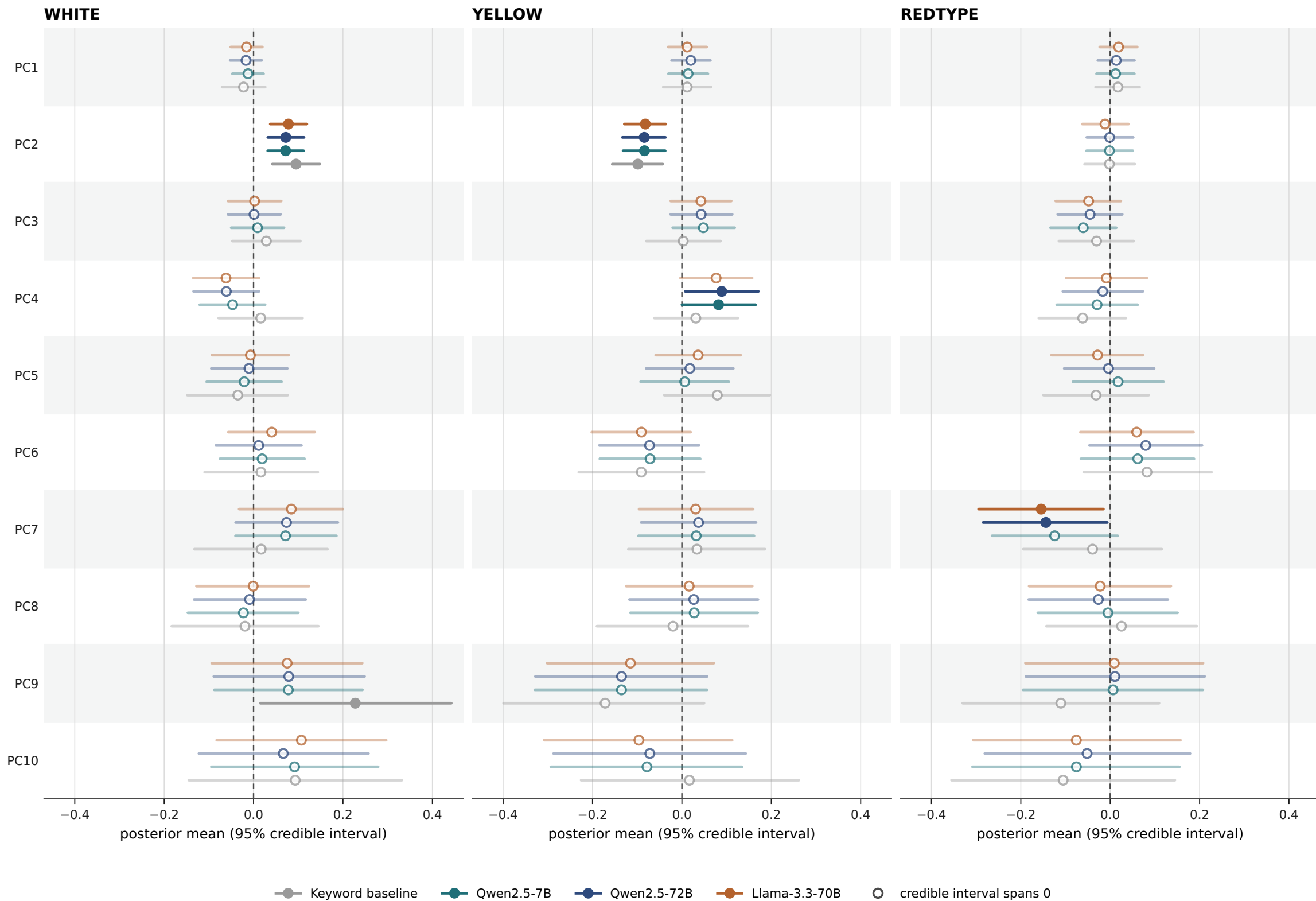


RQ4 Do the ecological associations survive a change of label source?



All 30 colour × PC effects per label source. MCMCglmm fixed effects, 3 chains per model (worst MPSRF 1.00053). Filled markers are significant at pMCMC < 0.05. n = 901 (Keyword baseline), 1,174 (Qwen2.5-7B), 1,166 (Qwen2.5-72B), 1,169 (Llama-3.3-70B); the keyword baseline labels fewer species, so its comparisons confound label accuracy with sample composition. Across all 120 tests only WHITE × PC2 and YELLOW × PC2 survive Benjamini-Hochberg correction ($q \leq 0.009$, significant under every label source); the 5 scattered marginal hits have $q \geq 0.36$ and are consistent with the ≈ 6 false positives expected at pMCMC < 0.05.